AgriLink

Digital Market Intelligence & Agricultural Value-Chain Platform

for Jimma Zone, Oromia Regional State, Ethiopia

Group -2

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1. Executive Summary

AgriLink is a proposed digital agricultural platform designed to connect smallholder farmers in Jimma Zone, Oromia Regional State, with real-time market intelligence, verified buyers, logistics providers, and (in later phases) financial services. Rather than functioning as a simple listings marketplace, AgriLink is designed as a decision-intelligence system: it tells a farmer not just what a crop is worth, but where selling it will generate the greatest net return once transport and logistics costs are factored in.

Ethiopian agriculture remains the backbone of the national economy, employing the majority of the workforce, yet it continues to be dominated by smallholder producers who face fragmented markets, thin price information, and limited digital tooling suited to their language and connectivity constraints. Government-backed efforts such as the 8028 Farmer Hotline, the 6077 market information hotline, the Ethiopian Commodity Exchange (ECX), and the newly launched OpenAgriNet initiative are closing parts of this gap at a national level. AgriLink is proposed as a complementary, hyper-local, Afaan Oromoo-first layer that goes deeper into a single zone than any national platform can, while remaining architecturally positioned to interoperate with those national systems rather than compete against them.

This proposal recommends against writing production code immediately. Instead, it defines a disciplined first phase of problem validation and technical design, followed by a phased build: a lean MVP focused on market intelligence and buyer matching for five priority crops in Jimma Zone, a supervised pilot with real farmers and buyers, and a data-backed decision on whether and how to scale toward Oromia-wide and eventually national coverage.

The remainder of this document sets out the problem, the competitive and policy context, the proposed solution architecture across seven system modules, a phased delivery plan, an indicative budget, and the risks that must be actively managed for AgriLink to succeed where narrower efforts have struggled.

2. Background and Problem Statement

2.1 Agriculture in Ethiopia and Oromia: Context

Agriculture remains central to the Ethiopian economy, accounting for roughly a third of national GDP and employing more than 60% of the workforce, the large majority of whom are smallholder farmers with limited access to reliable services and markets. Oromia is Ethiopia's largest and most agriculturally productive region, and average landholding sizes there (around 1.15 hectares) are already small and trending smaller as population density rises and land continues to be subdivided across generations.

Structural constraints recur across the research literature on Ethiopian smallholder agriculture: small and fragmented landholdings, climate variability and erratic rainfall, declining soil fertility, weak market integration, and low adoption of modern production and information technologies. A study of contract participation among farmers in central Oromia found that the majority of surveyed households (over 62%) relied on informal, undocumented sales arrangements, fewer than 5% had access to formal contracts, and roughly a third sold through open markets with no price protection at all a pattern consistent with low bargaining power and high exposure to local price shocks.

2.2 Core Problems Farmers Face, Production to Sale

Drawing on the research above together with the structure of Ethiopia's existing agricultural extension and market systems, the problem can be broken into five linked stages:

- **Information asymmetry at harvest.** A farmer typically knows the price at their nearest market, but not at markets 50–300 km away, and has no reliable way to weigh a higher price elsewhere against the transport cost of reaching it.
- **Thin extension coverage.** Ethiopia's extension worker-to-farmer ratio is estimated at roughly 1 to 200, so timely, personalized agronomic and market advice reaches only a fraction of farming households.
- **Fragmented buyer access.** Buyers (wholesalers, processors, cooperatives, exporters) and farmers largely find each other through informal networks and middlemen, which compresses farmer margins and slows price discovery.
- **Logistics as a black box.** Even when a better market is known, arranging transport, estimating its true cost, and trusting the transporter is a separate, largely undigitized problem.
- **Language and interface exclusion.** Most existing digital services are Amharic- or English-first; Afaan Oromoo speakers in a majority-Oromo zone are underserved by interfaces that do not match their first language.
- Transport and transit costs are unpredictable and partly informal. Farmers who do attempt to move produce toward better-paying markets face genuine transportation difficulty, along with transit fees incurred along the way that are not disclosed or standardized in advance, further eroding any price advantage a more distant market might otherwise offer.
- Finding a buyer is manual, slow, and unreliable, with real losses for perishable produce. Locating a merchant willing to buy is often described by farmers as tedious and time-consuming, and merchants themselves frequently cannot reach the farmer in time. For durable crops this causes delay and lost bargaining power; for short-shelf-life produce such as tomato and avocado, it routinely results in the harvest spoiling before a sale can be completed at all — a total loss rather than a discounted one.

2.3 Why Jimma Zone

Jimma Zone is one of Oromia's most productive agricultural areas and a sensible, bounded environment in which to validate AgriLink before wider rollout:

- **Crop diversity.** Coffee is the zone's dominant cash crop — recent estimates put zonal production above 65,000 tons a year, representing roughly a sixth of Oromia's total coffee output — alongside maize, teff, sorghum, wheat, pulses, enset, root crops, and horticulture, giving AgriLink a realistic multi-crop testbed.
- **Scale.** Jimma Zone contains several of the highest crop-producing woredas (districts) in the wider Oromia and SNNP production ranking, meaning a successful pilot would already touch a meaningful volume of national production.
- **Productivity gap.** Coffee yields in the zone remain low relative to potential (on the order of 7 quintals per hectare) due to aging trees and limited input use — a gap that better market signals and advisory access can help address over time.

- **Institutional anchor.** Jimma University and the zone's existing cooperative structures provide a natural base for field research, farmer trust-building, and pilot logistics

Table 1:Priority crop landscape in Jimma Zone

Crop	Role	Notes
Coffee	Primary cash crop	~65,500 t (2022/23), ~16.7% of Oromia's coffee output; smallholder plots average 0.5 ha
Maize	Leading food/cash cereal	Largest single contributor to food-crop production by volume in several woredas
Teff	Cash and staple crop	Important secondary cash source after coffee in most weredas
Sorghum, Wheat, Barley	Staple cereals	Grown across midland and highland agro-ecologies
Enset (false banana)	Food-security staple	Especially critical in highland weredas such as Setema and Sigimo
Pulses, root crops, horticulture	Diversification crops	Beans, peas, potato, beetroot, onion, mango, banana, tomato, pepper

3. Market and Competitive Landscape

AgriLink is not entering an empty field. Ethiopia has invested heavily in agricultural digitization over the past decade, led primarily by the Agricultural Transformation Institute (ATI) and the Ethiopian Commodity Exchange (ECX). Understanding what already exists, and precisely where it stops, is essential to defining AgriLink honestly rather than duplicating effort.

3.1 Government and Institutional Platforms

- **8028 Farmer Hotline :-** ATI's flagship IVR/SMS advisory service, launched in 2014, reaches an estimated 7 million registered users with agronomy and livestock guidance. It works on basic feature phones but is a one-way, generic advisory channel rather than a personalized market-matching tool.
- **6077 Market Information Hotline :-** Part of ATI's National Market Information System, delivering commodity price information across hundreds of markets to farmers, buyers, and policymakers by phone and web. It reports prices but does not compute net-of-transport recommendations or connect a specific farmer to a specific buyer.
- **Ethiopian Commodity Exchange (ECX) :-** Established in 2008, ECX provides regulated trading, grading, storage, and settlement for a defined set of commodities (notably coffee, sesame, maize, wheat), with prices pushed to electronic tickers, SMS, and a toll-free line, and an online trading platform launched in 2025. ECX is built around standardized, warehouse-graded lots and formal exchange members it is not designed for a farmer with a few thousand kilograms deciding between three nearby towns.
- **OpenAgriNet (Ethiopia OAN) :-** The most significant recent development: a national digital public infrastructure initiative launched by ATI in February 2026 with the Center

for Open Societal Systems and Protean eGov Technologies, backed by a contract valued at roughly USD 2.75 million. OpenAgriNet aims to unify farmer/farm identifiers, extension, market, climate, and financial data into a single interoperable, AI-enabled, voice-first network, integrated with Ethiopia's Fayda digital ID system and the national Digital Agriculture Roadmap (2025–2032). It is explicitly designed as infrastructure for other services to build on, not a single closed application.

- **Warehouse Receipt Systems :-** Ministry of Trade/IFC-backed programs that let farmers use stored crops as loan collateral, cross-referencing ATI market price data a useful precedent for AgriLink's future financial-services module.

3.2 Private and NGO-Led Tools

A smaller set of private and donor-backed mobile applications operate in adjacent spaces — for example BFARM-TECH, which connects Ethiopian smallholders with farm equipment operators and mechanics, and various NGO advisory tools. These tend to be narrow in scope (a single service line, such as equipment access) and are not zone-specific, Afaan Oromoo-native, decision-intelligence tools.

3.3 Gap Analysis: Where AgriLink Fits

The honest conclusion of this landscape review is that AgriLink should not be positioned as "Ethiopia's first digital agriculture platform." It should be positioned as the first hyper-local, Afaan Oromoo-first, transport-aware selling-decision tool for Jimma Zone — one that can consume national infrastructure (price feeds, and eventually OpenAgriNet's data rails and farmer IDs) rather than rebuild it, while owning the specific, high-value problem that no current platform solves end-to-end: telling an individual farmer, in their own language, exactly where to sell a specific harvest for the highest net return.

Table 3.1 : Competitive positioning summary

Platform	Category	Strengths	Gap vs. AgriLink
8028 Hotline	IVR/SMS advisory	Massive reach, feature-phone compatible	Generic advice, no market matching, one-way
6077 Hotline / NMIS	Price information	Broad market coverage, official price data	Raw prices only, no net-revenue logic or buyer link
ECX	Formal commodity exchange	Trusted settlement, graded quality, next-day payment	Built for bulk/graded lots, not smallholder daily decisions
OpenAgriNet	National digital infrastructure	AI-enabled, voice-first, ID-linked, well-funded	Nationwide scope, still early-stage, generalized by design
AgriLink (proposed)	Local decision intelligence + marketplace	Afaan Oromoo-first, transport-aware, buyer-matched, zone-deep	Limited initial geography; must earn trust and liquidity locally

4. Proposed Solution: AgriLink

4.1 Vision Statement

To become the digital agricultural infrastructure layer for Jimma Zone **connecting farmers with market intelligence, buyers, agricultural services, logistics, and finance** so they can make better decisions and capture more of the value their production already creates before expanding, on validated evidence, across Oromia and beyond.

4.2 Core Concept

A farmer registers their farm and crops; sees localized, comparative market prices; sees active buyers; requests transportation; receives agronomic and selling recommendations; sells; gets paid; and builds a transaction history that becomes an asset in its own right (for future credit scoring). A buyer posts demand with crop, quality, quantity, and location; finds matching farmers; negotiates or purchases; and arranges logistics. AgriLink sits between them, providing market intelligence, matching, price analytics, logistics coordination, payments facilitation, farmer records, and an Afaan Oromoo AI assistant.

4.3 Key Differentiators

- **Decision intelligence, not just listings:-** AgriLink's core output is a recommendation where to sell not a raw price table.
- **Afaan Oromoo-first design.** Voice and text interaction in the farmer's first language, not a translated afterthought.
- **Transport-aware economics.** Every price comparison nets out estimated transport cost and distance, because a nominally higher price 300 km away is frequently the worse choice.
- **Zone-deep, not country-wide.** Depth of buyer relationships, transporter relationships, and local price accuracy in one zone beats shallow national coverage for the MVP stage.
- **Built to interoperate, not duplicate.** Architected to consume ECX and OpenAgriNet data feeds and identifiers as they become available, rather than compete for the same national infrastructure budget.

5. Goals and Objectives

The project has one strategic goal and a set of measurable objectives for the three-week intensive development period, aligned to the phased plan in Section 12.

Strategic Goal

Validate, build, and pilot a decision-intelligence and market-linkage platform that measurably increases net revenue captured by smallholder farmers in Jimma Zone, as a foundation for a scalable agricultural technology venture.

Measurable Objectives (3-Week Intensive Development)

- Complete structured problem validation with at least 150 farmers and 30 buyers/traders across 3–5 woredas of Jimma Zone before writing production code.
- Ship an MVP covering Farmer Profile, Market Intelligence, Buyer Marketplace, and the "Where Should I Sell?" recommendation engine for 5 priority crops.
- Onboard at least 500 farmers and 50 verified buyers/transporters into the Jimma pilot.

- Demonstrate a measurable uplift in average net revenue per transaction for pilot farmers relative to a baseline group.
- Produce a validated, data-backed business case and technical roadmap for expansion beyond Jimma Zone.

6. Target Users and Stakeholders

6.1 Farmers (Primary Users)

Smallholder producers of maize, teff, wheat, potato, and coffee in Jimma Zone, predominantly Afaan Oromoo speakers, with mixed literacy and device access ranging from basic feature phones to entry-level Android smartphones. Their core need is a trustworthy answer to "where should I sell, and to whom?" delivered in a form that does not require reading complex dashboards.

6.2 Buyers

Wholesalers, processors, retailers, cooperatives, exporters, and restaurants who need predictable, quality-specified supply and currently rely on informal broker networks with limited visibility into what is actually available, where, and when.

6.3 Transporters

Truck and pickup owner-operators who move produce between rural collection points and urban or secondary markets, currently found through word of mouth, with no structured way to advertise capacity or receive matched transport requests.

6.4 Institutional Stakeholders

- **Cooperatives and unions :-** potential aggregation points for farmer onboarding and bulk logistics.
- **Jimma University and local government/Bureau of Agriculture :-** potential research, validation, and extension partners.
- **ATI / OpenAgriNet and ECX :-**potential long-term data and infrastructure partners rather than competitors.
- **NGOs and donors active in the zone :-** potential co-funders of the pilot and monitoring & evaluation work.

7. Scope of Work: System Modules

AgriLink is designed as seven modules.

7.1 Module 1 :- Farmer Profile & Farm Registry

- Farmer registration (name, phone/ID, location, cooperative membership)
- Farm records: land size, GPS location, crops grown, harvest periods, production capacity
- Transaction history, building toward a future creditworthiness signal

7.2 Module 2 :- Market Intelligence Engine

- Current and historical prices by crop, market, and quality grade

- Price trend visualization and simple short-term forecasting
- Cross-market comparison, feeding directly into the "Where Should I Sell?" engine (Section 8)

7.3 Module 3 :- Buyer Marketplace

Buyer demand posting: crop, quality, quantity, location, price offered

Search and matching between farmer supply and buyer demand

In-app negotiation/agreement flow and buyer verification

- Multi-tier buyer visibility. Distinguish local collector-merchants, who buy in-village and transport produce onward to Jimma city, from large city merchants who pay better but are harder to reach directly — so a farmer can see both tiers and judge, for a given harvest, whether reaching the city buyer directly is realistic or whether working with a collector is still the better option that trip.
- Live "buying now" status. Buyers and merchants signal real-time availability and location rather than posting a static, one-off demand listing — farmers today report that simply finding a merchant who is actually buying, on the day they need to sell, is one of the most time-consuming parts of the process.
- Merchant reliability signals. Track response time and no-show history. Farmers report merchants failing to arrive on time and produce going to waste as a direct result; a visible reliability record lets a farmer judge who to trust before they wait on someone who will not come.

7.4 Module 4 :- AgriLogistics

Transporter registry with vehicle type, capacity, and coverage area

Transport request creation and route/price matching

Basic shipment status tracking (requested → confirmed → in transit → delivered)

- Transit fee transparency. A maintained log of typical transport costs and the informal transit/checkpoint fees farmers report paying along common routes into Jimma city, so the true landed cost of reaching a given market is visible before a farmer commits, rather than discovered painfully along the way.
- Verified collector role. Let today's small itinerant merchants — who currently buy locally at a low price and resell into Jimma city at a markup — register as a legitimate, verified collection-and-transport partner with a disclosed, transparent fee, rather than treating them purely as a markup to be routed around. Many are already doing real transport and market-access work that is not realistic to eliminate everywhere, particularly for farmers furthest from the city.

7.5 Module 5 :- AgriAI (Deferred to Phase 2/3)

- Afaan Oromoo conversational assistant for price queries and selling recommendations
- Farming calendar and basic agronomic advice
- Crop disease identification (later-stage, image-based) once sufficient local training data exists

7.6 Module 6 :- Financial Services (Explicitly Out of MVP Scope)

Digital payments, savings, agricultural insurance, credit scoring, and input financing require regulatory engagement (payment service provider licensing, partnerships with banks/microfinance institutions) that sits outside a software build. This module is deliberately

excluded from the MVP and pilot, and is scoped only as a future-phase initiative contingent on the right partnerships.

7.7 Module 7 — Agricultural Intelligence Dashboard (Deferred)

An aggregated view for cooperatives, government, and NGOs registered farmers, hectares under cultivation, expected and available harvest volumes, average prices, and regional supply/demand signals. Valuable, but dependent on Modules 1–3 reaching sufficient data density first.

8. Flagship Feature Deep-Dive: "Where Should I Sell?"

This is the feature intended to differentiate AgriLink from every existing price-information tool in the market. Instead of showing raw prices per market, AgriLink computes an estimated net revenue per market by combining the local price, the estimated transport cost for the farmer's actual volume and location, and the distance/route involved, then ranks markets by what the farmer would actually take home.

Table 2:**Worked example : 5,000 kg of potatoes**

Market	Price (ETB/kg)	Distance	Est. Transport Cost (ETB)	Est. Net Revenue (ETB)
Jimma	25	20 km	3,000	122,000
Nekemte	28	150 km	12,000	128,000
Addis Ababa	32	350 km	35,000	125,000

Naively, Addis Ababa's price of 32 ETB/kg looks best. Once transport is netted out, **Nekemte** produces the highest net revenue (ETB 128,000), which is the recommendation AgriLink surfaces along with the estimated transport cost, so the farmer can verify the logic rather than simply trust a black box.

Delivery in Afaan Oromoo

The same interaction is designed to work conversationally. A farmer can ask, in Afaan Oromoo, how many kilograms of wheat they have and where they should sell today; AgriLink responds with a market comparison and a recommendation, and can continue the conversation (for example, surfacing which buyers are currently active in the recommended market).

Underlying requirements

- A maintained, geocoded database of markets and reference prices per crop, refreshed regularly (initially through partner data feeds and manual/field data collection, later through ECX/OpenAgriNet integration where available).
- A routing and transport-cost estimation layer (initially rule-of-thumb ETB/km/ton by route type, refined over time using AgriLogistics transaction data).
- A conversational interface in Afaan Oromoo, built on a large language model with a curated agricultural knowledge layer and strict guardrails against fabricated prices.

9. Geographic and Product Rollout Strategy

The most important strategic decision in this proposal is what NOT to build first. AgriLink will not attempt Oromia-wide or national coverage at launch. It will start narrow, prove the model, and expand only on evidence.

Stage 1 Jimma Zone

- Farmers: Jimma Zone, focused woredas with strong crop production and cooperative presence
- Crops: maize, teff, wheat, potato, coffee,tommato, and carrot
- Markets: Jimma town plus 2–4 nearby major secondary markets (e.g., Nekemte and Agaro as reference points, to be confirmed in Phase 0 field research)
- Users: farmers, buyers, and transporters within and immediately around the zone

A note on the Stage 1 crop list: the seven crops above (maize, teff, wheat, potato, coffee) are all storable commodities, selected on production volume and market depth. Field observation in Jimma Zone (Section 2.2) indicates that some of the most acute farmer losses occur with short-shelf-life produce such as tomato and avocado, where a delayed sale results in total spoilage rather than a lower price. This list should therefore be treated as provisional rather than final: Phase 0 (Section 12.3) should explicitly test whether one or two perishable crops belong in the initial five, or whether they are better sequenced as a fast-follow once the perishability and time-sensitivity layer (Section 7.5) is built and proven on storable crops first. Either way, the decision should be made deliberately and recorded before Phase 1 development scope is locked.

Stage 2 Oromia Expansion

Only pursued if Stage 1 demonstrates retained usage, real transactions, and a measurable revenue uplift for farmers. Expansion would prioritize adjacent zones with similar crop profiles and existing cooperative infrastructure.

Stage 3 National

A national push would depend on both Stage 1/2 evidence and the maturity of OpenAgriNet, since national-scale identity, price-feed, and interoperability infrastructure is exactly what that initiative is building; AgriLink's long-run advantage at national scale would come from integrating with it rather than rebuilding it.

10. Technical Architecture

10.1 Technology Stack

Table 3:Tech to be used

Layer	Choice	Rationale
Web	Next.js (React)	Server-side rendering for low-end connections; shared component logic with mobile
Mobile	React Native / Expo	Single codebase for Android-first

Layer	Choice	Rationale
		distribution (dominant OS among target users)
Backend	NestJS (Node.js)	Modular, testable service architecture aligned to the 7 system modules
Database	MYSQL	Relational integrity for transactions; geospatial queries for markets/distance/routing
Maps & Geospatial	Mapbox (or OpenStreetMap-based alternative)	Distance/route estimation for the net-revenue engine
AI / Conversational Layer	LLM API + curated agricultural knowledge base	Afaan Oromoo assistant; retrieval-grounded to avoid fabricated prices
Payments (future)	Licensed Ethiopian payment/mobile-money providers	Deferred to Module 6; requires partnership and regulatory work
Notifications	SMS gateway + push notifications	SMS as the reliable fallback for feature-phone users
Infrastructure	Cloud hosting with a regional/CDN edge	Optimized for low-bandwidth delivery; see 10.4

10.2 System Architecture Overview

AgriLink follows a modular service-oriented architecture: a set of backend services (Farmer, Market Intelligence, Marketplace, Logistics, AI Assistant) sitting behind a shared API gateway, each owning its own data but sharing a common farmer/farm identity model. This mirrors the seven-module scope directly, so that Modules 5–7 can be added later without re-architecting Modules 1–4, and so that individual services can eventually be swapped for or federated with national infrastructure (for example, resolving farmer identity against OpenAgriNet's ID system, or reading reference prices from ECX/6077 feeds) without a full rebuild.

10.3 Data Model Overview (Core Entities)

- Farmer → Farm(s) → Crop Records (crop, area, expected yield, harvest window)
- Market → Price Observation (crop, grade, price, date, source)
- Buyer → Demand Posting (crop, grade, quantity, location, offer price)
- Transport Provider → Transport Request → Shipment
- Transaction (links Farmer, Buyer, Market, optional Shipment, final price, status)

10.4 Connectivity and Accessibility Strategy

Design decisions here are driven directly by the connectivity reality of the target users, not by convenience for the development team. National data shows internet penetration in Ethiopia around 22% of the population, with over three-quarters of the population living in rural areas where connectivity and device access are more constrained; by contrast, mobile money adoption has scaled to well over 100 million registered accounts, driven substantially by USSD, which runs over basic 2G connections without requiring a smartphone or data plan.

- **Tiered access channels:** full app experience for smartphone/data users; a lightweight, low-bandwidth web view for intermittent connectivity; SMS and USSD-style menus and the 8028-style IVR pattern for feature-phone users, so the core "where should I sell" answer is reachable without a smartphone.
- **Offline-tolerant mobile app:** local caching of recent prices and farm records, with background sync when connectivity returns.
- **Voice-first Afaan Oromoo interaction** as a first-class channel, not an add-on, consistent with literacy and language realities in the target user base.

10.5 Integration Strategy

Rather than treating ECX, the 6077 hotline, and OpenAgriNet as competitors to out-build, AgriLink's architecture treats them as potential upstream data and identity providers. Concretely: reference price feeds can be validated against 6077/ECX published data where available; farmer identity can, over time, be reconciled with OpenAgriNet's farmer/farm identifiers once that infrastructure is generally available for third-party integration; and AgriLink's own zone-level data (hyper-local prices, buyer activity, logistics costs) is a genuine contribution back to that wider ecosystem rather than a duplicate of it.

11. Development Methodology

The project follows a research-first, phased-agile approach: nothing is built before the underlying assumption is validated with real farmers and buyers, and each subsequent phase is gated on evidence from the previous one.

- **Phase 0 :- Discovery:** desk research, field interviews, and competitive/regulatory analysis before any design or code.
- Phase 1 :- Design & MVP Build: rapid intensive development following prototype validation, with parallel frontend, backend, data, and QA work to deliver the MVP within the three-week schedule.
- Phase 2 :- Guided Pilot: supervised rollout and rapid feedback cycles during the final development week, with continuous testing and fixes.
- Phase 3 :- Evaluation: structured end-of-project measurement against the KPIs, culminating in a go/no-go, evidence-based case for the next stage.

Throughout, the project maintains a single, versioned backlog aligned to the seven modules in Section 7, so scope creep into Modules 5–7 during the MVP phase is a visible, deliberate decision rather than an accident.

12. Project Plan and Timeline

The project will be completed over 3 intensive weeks, using a highly focused and parallel development approach. The four accelerated phases are research and validation, MVP development, pilot testing, and evaluation, with activities overlapping where necessary to meet the compressed schedule.

Phase 0 :- Research & Validation | Week 1

Goal: Quickly validate the problem and define the MVP.

- Interview farmers and buyers.
- Review competitors and existing solutions.
- Identify priority crops and markets.
- Map key farmer challenges.
- Finalize MVP requirements and technical architecture.
- Create and test a basic prototype.

Deliverables: Validated problem brief, MVP requirements, prototype, and go/no-go decision.

Phase 1 — MVP Development | Week 2

Goal: Build a functional MVP based on the validated requirements.

- Develop the core Farmer Profile and Market Intelligence features.
- Build the Buyer Marketplace and AgriLogistics features.
- Implement the **"Where Should I Sell?"** recommendation concept.
- Develop the main web/mobile interface.
- Add basic market and price data.
- Conduct initial testing and security checks.

Deliverables: Functional MVP, initial database, and testing report.

Phase 2 — Pilot Testing | Week 3

Goal: Test the MVP with a small group of real users.

- Onboard a selected group of farmers and buyers.
- Test the main features and user experience.
- Collect feedback and identify problems.
- Test market-price information and recommendations.
- Fix critical issues and improve the MVP.

Deliverables: Tested MVP, user feedback, pilot results, and initial case studies.

Phase 3 — Evaluation & Next Steps | End of Week 3

Goal: Evaluate the results and determine the next stage.

- Analyze user feedback and system performance.
- Review the MVP against the original objectives.
- Identify improvements and remaining challenges.

- Prepare a short evaluation report.
- Develop a roadmap for future expansion.

Deliverables: Evaluation report, improved MVP, and scale-up roadmap

13. Risk Analysis and Mitigation

Table 4:Risk analysis and mitigation

Risk	Likelihood	Impact	Mitigation
Farmer trust / adoption is slower than expected	High	High	Lead with cooperative partnerships and in-person onboarding; keep the SMS/voice channel as a low-friction entry point; do not require a smartphone for core value.
Price data becomes stale or inaccurate	Medium	High	Combine field-collected prices with official feeds (6077/ECX) where available; show data recency to users; build a lightweight field-verification workflow.
OpenAgriNet or a similar national platform absorbs the market before AgriLink scales	Medium	High	Design for interoperability from day one (Section 10.5); position AgriLink as a zone-deep complement, not a national competitor, and pursue partnership conversations early.
Connectivity/device limitations exclude target users	High	Medium	USSD/SMS/IVR fallback channel is a first-class part of the MVP, not a later add-on.
Afaan Oromoo AI assistant gives inaccurate or fabricated advice	Medium	High	Ground all AI responses in verified structured data (prices, distances); restrict free-form generation around numeric claims; human review of content before Phase 2.
Buyer-side liquidity is too thin to make the marketplace useful	Medium	High	Actively recruit and onboard buyers in Phase 0/1, not just farmers; consider seeding early demand through cooperative and processor partnerships.
Regulatory complexity delays financial features	Low (Year 1)	Medium	Module 6 is explicitly excluded from MVP/pilot scope; regulatory engagement runs in parallel, not on the critical path.
Team/funding constraints slow delivery	Medium	Medium	Phase 0 gate exists precisely to confirm team and budget realism before Phase 1 commitments are made.

Risk: Existing informal merchants resist the verified collector model

Likelihood: Medium

Impact: Medium

Mitigation: Today's small itinerant merchants earn their margin from an undisclosed markup between the village price and the city price; asking them to register and operate on a transparent, disclosed fee is a direct change to their current business model, and some may decline or work around the platform rather than adopt it. Mitigate by making the collector role genuinely beneficial rather than purely restrictive — for example, faster access to farmers ready to sell, reduced time spent searching for supply, and a reliability record that helps trustworthy collectors win more business — so that participation is a net gain for cooperative merchants, not just a constraint. Treat collector adoption as its own tracked outcome in the Phase 2 pilot rather than assuming it will follow automatically from farmer adoption.

14. Success Metrics and KPIs

Metric	Target
Registered farmers (Jimma pilot)	≥ 500 by end of Phase 2
Registered buyers / transporters	≥ 50 verified accounts by end of Phase 2
Completed transactions facilitated through AgriLink	≥ 300 by end of Phase 2
Average net revenue uplift vs. baseline (non-users)	Measurable, statistically supported uplift reported in Phase 3
“Where Should I Sell?” queries per active user / month	Tracked as core engagement signal

Metric	Target
SMS/USSD channel usage share	Tracked to confirm reach beyond smartphone users
Farmer-reported trust/satisfaction (survey)	≥ 70% positive in Phase 3 evaluation survey
Data freshness (avg. age of active price records)	≤ 7 days for priority crops in pilot markets
Spoilage/waste reduction on perishable listings	Measurable reduction in farmer-reported spoilage loss vs. baseline, tracked once perishable crops are active
Merchant/buyer reliability (on-time arrival, no-show rate)	≥ 80% on-time fulfillment rate among verified collectors and buyers by end of Phase 2
Verified collector adoption rate	≥ 50% of active local collector-merchants in pilot woredas registered under the transparent-fee model by end of Phase 2

15. Business Model and Sustainability

AgriLink's pilot phase is intentionally treated as a validation and social-impact exercise, not a revenue-generation exercise: farmer- and buyer-facing core features (price intelligence, matching) should remain free or near-free during Phase 2 to maximize adoption and data quality. Longer-term sustainability options to be evaluated against pilot evidence include:

- **Transaction/logistics fees :** a small facilitation fee on completed marketplace transactions or matched logistics, modeled on ECX's settlement role.
- **Buyer-side subscriptions :**premium demand-visibility or analytics features for wholesalers, processors, and exporters, who have higher willingness to pay than individual farmers.
- **Institutional/NGO/government contracts :** the aggregated intelligence dashboard (Module 7) as a paid or grant-funded service for cooperatives, government bureaus, and donors.
- **Financial-services partnerships (Future Phase) : referral or revenue-share arrangements with licensed microfinance/payment providers once Module 6 is pursued, rather than AgriLink becoming a licensed financial entity itself.**

The specific mix and pricing will be determined in Phase 3, informed directly by what pilot farmers and buyers demonstrate they are willing to pay for : research on rural information services in Ethiopia suggests some willingness to pay for reliable price information, but this should be tested locally rather than assumed.

16. Conclusion and Next Steps

AgriLink addresses a well-documented, high-stakes problem fragmented market access for smallholder farmers with a specific, defensible angle: transport-aware, Afaan Oromoo-first selling recommendations, delivered zone-deep in Jimma before any attempt at broader scale. Ethiopia's existing digital agriculture ecosystem, from the 8028 hotline through to the newly launched OpenAgriNet initiative, shows both that this problem is taken seriously at a national level and that a real gap remains at the hyper-local, decision-support layer that AgriLink is designed to fill.

Appendix A: Research Phase 1 Work Plan

The following twelve tasks constitute the Phase 0 discovery work referenced throughout this proposal, and are the recommended immediate starting point rather than any production coding:

- Identify the 10 biggest agricultural problems in Oromia, ranked by farmer-reported severity.
- Map the exact problems farmers face from production through harvest to sale, stage by stage.
- Research existing Ethiopian agricultural platforms in depth (8028, 6077, ECX, OpenAgriNet, BFARM-TECH, and others) including direct outreach where possible.
- Define precisely what AgriLink does differently from each, in writing, before design begins.
- Confirm the first 5 priority crops with local agronomic and market advisors.
- Conduct structured interviews and/or surveys with farmers and buyers to validate the problem and the proposed solution.
- Finalize the technical architecture in Section 10 against real infrastructure and connectivity findings from the field.
- Build a clickable prototype and test it with a small panel of farmers and buyers before backend development starts.
- Plan the Jimma pilot logistics: onboarding events, field coordinator recruitment, and cooperative partnerships.

Appendix B: Sources and References

Research informing this proposal drew on the following categories of sources; full citations should be retained in the project's research repository:

- Peer-reviewed and preprint research on Ethiopian smallholder market participation, commercialization, and agricultural information access (including studies on central and southwestern Oromia).
- World Bank and FAO reporting on Ethiopian agricultural development programs and digital agriculture initiatives.

- Ethiopian Agricultural Transformation Institute (ATI) public communications on the 8028 Hotline, the 6077/National Market Information System, and the OpenAgriNet initiative (launched February 2026).
- Ethiopian Commodity Exchange (ECX) public information on trading, settlement, and market-data dissemination.
- GSMA, DataReportal, and Internet Society reporting on Ethiopian mobile, internet, and mobile-money penetration.
- Regional/zonal agricultural profiles and field assessments covering Jimma Zone crop production and livelihoods.